FINTECHATHON 2026 | INTERNATIONAL TRACK | TOPIC C

StudyRoute

Execution Evidence Report

What we ran, what happened, and where the raw files are: a full regression run, a live local API demonstration, an audit tamper-and-restore experiment and ledger checks, all on synthetic data.

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REPOSITORY

https://github.com/NoobCrow/StudyRoute

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01 / SUMMARY

Summary

This report shows what happened when we ran StudyRoute on a clean machine and drove it end to end. Everything is synthetic: the providers, the KYC results, the rails and the settlements are all simulated, no real money moved and no live bank was contacted.

We ran the full backend test suite, played the whole 14-step payment demonstration against a real local API, edited an audit record on purpose to see the chain break, and checked the bank ledger and the AI components. The table below is the short version. Each row names the evidence file behind it (E01 to E07, listed under Repeating it yourself).

What we checked	Result	Evidence
Full backend regression	1,672 passed, 0 failed, 1 skipped (1,673 collected)	E01
Local API demonstration	14 steps, 66 checks OK, 0 failed	E02
Audit chain, untouched	PASS, 76 events	E03
Audit chain after we edited event 75	FAIL at event 75; PASS again after restoring it	E03
Chain recomputed offline, outside the app	Valid; the altered copy is rejected	E05
Bank ledger	Balanced in all 7 currencies	E04
AI components	Reader, dialogue policy, risk model and local explainer all load	E06
Repository scan for secrets and TODOs	Clean	E06

02 / WHAT WE RAN IT ON

What we ran it on

We took the source archive of commit bcce930 of NoobCrow/StudyRoute (834 files, SHA-256 4e6ce74c…74134a) and ran it on 18 October 2026 between 10:07 and 10:08 UTC.

Item	Setting
System	Linux, Python 3.13.14, pytest 9.1.1
App	FastAPI 0.143.0, SQLAlchemy 2.1.4, a throwaway SQLite database
Local AI	ONNX Runtime 1.31.0, sentencepiece 0.2.2
Network	API bound to 127.0.0.1:18763 only
Settings	sandbox mode, all four model modes on auto, cloud assistant off

The demonstration used the real application, worker, database and local models. Simulated providers sent signed callbacks into the local API. The backend tests use in-process clients and fake clocks, and many of them turn the models off so they can check the deterministic rules on their own.

We did not do a clean dependency install, a frontend build, a browser run, a Docker run or any model training for this report. The health field `ai_mode=llm` means the local trained explainer is in use, not a remote model call; we confirmed that with a separate load check.

03 / BACKEND TESTS

Backend tests

We ran the complete backend suite without changing any application code: **1,672 passed, 0 failed, 0 errors, 1 skipped**, out of 1,673 collected, in 375.54 seconds. The per-test results are in the JUnit file (E01).

```
cd StudyRoute-main/backend
python -m pytest -o addopts='' -q --junitxml=../../validation/backend-current.xml
```

The skipped item was a whole module, `test_explainer_model.py`. Its data file, `backend/tests/data/explainer_cases.json`, was missing from the archive because the repository's `.gitignore` rule for `data/` had excluded it. So its quality-floor tests did not run that day. We fixed this on 19 October by committing the file, and a full run on our Windows machine now collects 1,705 tests, 1,704 passing and 1 skipped (a symlink test), including a new 25-prompt assistant suite. That later run is not part of the 18 October execution and its output is not in the evidence bundle. The 1,705 figure is the one used in the technical documentation, the deck and the README.

The suite covers sign-in and sessions, roles and household isolation, payment guards and idempotency, signed provider webhooks, the audit chain, document encryption at rest, the bank ledger and review desk, escrow, and the AI number guard and red-team checks. It shows that these assertions hold; it is not a claim of complete coverage.

04 / THE 14-STEP DEMONSTRATION

The 14-step demonstration

The `export_evidence` runner drives the real local API and prints one OK or FAIL line per assertion. The whole flow gave 66 OK and no failures (E02). The table is an index to that transcript.

Step	What happens	What we saw
1	Dashboard and deadline	A synthetic tuition goal and deadline are there
2	Compare providers	Three synthetic providers ranked by total cost, not a live quote
3	Explain the comparison	The assistant answers from the engine's facts
4	Sentence to plan	Planning stays separate from sending; a human decision is still needed
5	Draft and safe retry	The same idempotency key returns the same payment, not a second one
6	KYC gate	The request is blocked until the mock identity check is done
7	Prepare to authorise	Mock KYC reviewed; recipient allowlisted with the step-up code
8	Approve	Student and wrong-code attempts refused; parent with a valid code accepted; replay is harmless
9	Hostile inputs	A forged webhook is rejected; a prompt-injection instruction is refused
10	Tracker	The payment moves through submission to simulated settlement
11	Compliance report	A report with an integrity hash is produced (not a legal attestation)
12	Bank network	Synthetic national and international transfers, a return, and balancing entries
13	Smart escrow	Tranche and freeze behaviour on a local simulated EVM, no real tokens
14	Audit chain	Verification passes; the tamper experiment is exported separately

The exported payment is a synthetic GBP 10,000.00 tuition payment through Provider B, ending in SETTLED with a UETR-style tracking id. Ids and timestamps change on every run, so the raw JSON is the authority (E04).

Two things to keep in mind. SETTLED here means a simulated provider's callback moved our own state machine; it says nothing about a bank transfer or SWIFT delivery. And the sandbox MFA code and mock KYC are not meant for real identity checks.

05 / AUDIT TAMPER EXPERIMENT

Audit tamper experiment

Every event in the audit log stores the hash of the one before it. To see that this really works, we changed one stored amount on a throwaway database and asked the app to verify the chain.

Stage	Result	File
Before the edit	PASS, 76 events	audit_verify.json
Edit amount_minor of event 75	FAIL, HASH_MISMATCH at event 75	tamper_demo.json
Restore the original row	PASS, 76 events	tamper_demo.json
Offline recomputation of the fresh chain	Valid; altered copy rejected at 75	evidence-integrity-check.json
Offline recomputation of the chain that came with the archive	75 events valid; altered copy rejected at 74	evidence-integrity-check.json

The audit head after the run was `0e0ff18d559f93c6003fefe7cbb389e5037d93a06540687f372754461137678c`.

One detail cost us time: the API export adds a display field, `amount_display`, that is not stored and is not hashed. The hash covers only the stored fields in `HASHED_FIELDS`, so the offline check has to leave out `amount_display` and `this_event_hash`. If you include them you get a false mismatch.

There is also a limit we want to state plainly. Someone who owns the database and rewrites both the audit log and the simulated chain can still pass the local checks. A checkpoint kept somewhere separate will expose a changed prefix, and the test suite has a test for exactly this, but it is not the same as external immutability. Only run the tamper script on disposable data.

06 / LEDGER & ESCROW

Ledger and escrow

The bank network snapshot reports `all_ok=true`. Debits equal credits in all seven currencies, the mirror accounts are reconciled and the mismatch list is empty (E04). These are consistency checks on invented accounts.

Currency	Debits (minor units)	Credits (minor units)
AUD	8,050,000,000	8,050,000,000
BDT	61,604,821,000	61,604,821,000
CAD	8,050,000,000	8,050,000,000
EUR	8,050,000,000	8,050,000,000
GBP	50,120,980,000	50,120,980,000
MYR	8,050,000,000	8,050,000,000

Currency	Debits (minor units)	Credits (minor units)
USD	8,050,000,000	8,050,000,000

Transfer	Final state	Journal records
National	SETTLED	4
International	SETTLED	4
International	RETURNED	6

The export has one escrow with its contract-call records and one audit anchor. The simulated chain reports ready and the anchor comparison passes. It runs on py-evm, an in-memory EVM with no peers and no real tokens. Balanced books do not prove the transactions are realistic, and none of this validates real rail limits, fees or a production contract.

07 / THE AI COMPONENTS

The AI components

On this run we confirmed that the Reader, DialoguePolicy, RiskScorer and the local explainer load and take part in the live demonstration with the cloud path off (E06). We did not retrain anything or run a new accuracy study. The figures below come from our own evaluation files and were not re-run for this report.

Measure	Value	How to read it
Reader, blind routing	100 of 113 (88.50%)	A frozen holdout we measured earlier
Reader, attack blocking	13 of 14 blind, 18 of 20 hard	Small sets, and some attacks get through, so authorisation never depends on the filter
Dialogue policy, wrong plans	50 in 6,000	Before runtime review
Ask-everything-and-confirm baseline	1 in 6,000	The learned policy is not safer than this baseline in our simulator
Risk desk, medium and high	Precision 29.22%, recall 56.48%	Synthetic data; it routes payments to a person, it is not a fraud detector
Explainer, output accepted	80.86% of 700 pairs	Rewrites that fail the checks fall back to the engine's text

None of these numbers give the AI any power to approve; approval stays with the deterministic rules and a human with a fresh MFA code.

08 / EARLIER RESULTS & CHECKS MADE LATER

Earlier results and checks made later

These records came with the source and we did not re-run them.

Record	Result	Note
release/backend.xml	1,542 cases, no failures	An earlier full run; overlaps with the current one
release/backend-final-regression.xml	42 cases, no failures	Focused wording and review checks

Record	Result	Note
release/packaging-recheck.xml	53 cases, no failures	Packaging and preflight
release/web-tests.log	439 web unit and component cases	Taken before the bank-review screens' tests were merged; the merged tree has 463 (docs/TEST_CASES.md)
release/browser-tests.log	24 browser smoke tests	Static UI with a mocked API
security_2026-10-18/	146/146 attacks, 8/8 abuse, 42/42 assistant, 14 MITM defended, 48/48 browser	Our own probes, not an independent pentest
release/demo-recording.json	5 scenes, 95-second recording of the real local UI and API	Not the full MFA-to-settlement path

We do not add these counts together; the 18 October JUnit file (E01) is the current backend result.

After the run we also checked the repository itself (E07). The remote main branch is commit bcce930, the one this report covers. The repository is public and its README, technical documentation PDF and deck load without signing in. GitHub shows the "ci" workflow on that commit as successful (run 39081290614): docker, contracts, web, secret scan, backend and dependency audit. We did not read the job logs again.

Separately, in an Ubuntu 24.04 VM with Docker 29.1.3 we ran the smoke checks, the 14-step demo (66 checks OK), the bank-review demo, a real-Chrome audit (48 of 48) and the hardened overlay with a read-only filesystem and all capabilities dropped. They passed. That run used the archive of commit 8d91cc8 with the files of fix c9d044b copied in, so it was not a fresh archive of bcce930, and we did not repeat it for this report. It first showed that the trained explainer was rewording the assistant's questions, which made three demo checks fail; the fix is in bcce930. The raw output is in inherited/submission/evidence/docker_2026-10-18/.

09 / REPEATING IT YOURSELF

Repeating it yourself

Use a separate workspace and synthetic data only.

```
cd StudyRoute-main/backend
python -m venv .venv
python -m pip install -r requirements-dev.txt
python -m pytest -o addopts='' -q --junitxml=backend-current.xml
```

For the live demonstration, set a new `DATA_DIR` and `DATABASE_URL` outside the source tree, with `APP_ENV=sandbox`, `WEBHOOK_BASE_URL=http://127.0.0.1:18763`, all four model modes on auto and no cloud keys. Then:

```
python -m app.scripts.seed --if-empty
python -m uvicorn app.main:create_app --factory --host 127.0.0.1 --port 18763 --no-access-log

# in a second terminal, same environment
python -m app.scripts.export_evidence --base-url http://127.0.0.1:18763 --out /path/to/new-evidence --
tamper
python scripts/scan_repo.py
```

The `--tamper` option edits an audit row and restores it, so it belongs on disposable data only. Never use it, or `--reset`, on a deployment. The helper scripts we used (`tools/run_live.py`, `tools/inspect_evidence.py`) have the paths of our workspace in them; change those before reusing them.

Where the files are

ID	What it is	Files
E01	Full backend regression	validation/backend-current.log, .xml, .exit
E02	Live API demonstration	validation/live-run.json, live-export.log, live-evidence/sandbox_operation_log.txt
E03	Audit and tamper evidence	live-evidence/audit_log.json, audit_verify.json, tamper_demo.json
E04	Payments, ledger and escrow	live-evidence/payments.json, bank_network.json, smart_escrow.json
E05	Source identity and offline chain check	validation/source-identity.json, source-file-inventory.json, evidence-integrity-check.json
E06	Source scan and model loading	validation/source-scan.log, environment-models.json, rbac-matrix.json
E07	Remote repository and CI check	validation/final-verification.json

`SHA256SUMS.txt` in the evidence pack lets you check the files are unchanged. That only shows they match the list; it is not a signature. The database, generated keys and `dev_secrets.json` are deliberately left out.

10 / WHAT THIS DOES & DOES NOT SHOW

What this does and does not show

It shows that the source we submitted runs, passes its own tests, and behaves as described in a synthetic local demonstration, including the approval rules and the tamper-evident audit log. It does not show a public deployment, an independent security review, real-world model accuracy, regulatory approval or a live bank, FX or KYC connection. For the open risks, read the Security Self-Assessment. We have not sent the submission e-mail or given anyone access to the repository as part of this report.